

Windows Security Assessment Report

Al Shad Real Estate - Fictional Laboratory Case Study

Portfolio purpose: Demonstrate practical Windows security assessment, evidence handling, PowerShell automation and risk-based reporting in a controlled laboratory environment.

1. Executive Summary

A read-only Windows endpoint security assessment was executed using Windows PowerShell 5.1. The toolkit was independently validated as 10 modules with exactly 50 exported audit functions, then run against one Windows 10 laboratory endpoint. The run completed and generated structured CSV, JSON and HTML outputs.

The assessment identified configuration areas requiring review across privileged access, host firewall configuration, disk encryption, SMB hardening and Windows audit-policy coverage. Endpoint-protection and selected event-log checks included collection limitations; these are explicitly distinguished from confirmed security findings.

Item	Observed result
Platform	Windows 10 Home, Build 19045, 64-bit
PowerShell	5.1.19041.2364
Toolkit	10 modules / 50 exported audit functions
Checks executed	50
PASS	12
WARN	8
INFO	22
ERROR / collection limitation	8

Key message for reviewers: the project demonstrates an evidence-first workflow rather than claiming real client work, ISO certification, or penetration-testing experience.

2. Scope and Methodology

Scope was intentionally limited to one Windows 10 endpoint. The toolkit uses native Windows and PowerShell interfaces and performs read-only assessment actions. No exploitation, password changes, firewall modification or destructive configuration changes were used.

Audit areas included system baseline, local accounts, RBAC, Security Event Logs, endpoint-protection visibility, Windows Firewall, network exposure, services, hardening controls and security-policy evidence. Each module exports five checks for a total of 50 functions.

ISO/IEC 27001:2022 treatment: references are used only as control/evidence-mapping themes. This laboratory exercise is not an ISO certification, conformity assessment or organisational ISMS audit.

3. Verification Evidence

Verification item	Observed result
PowerShell version	5.1.19041.2364
64-bit process	True
Windows build	19045
Module files	10
Exported audit functions	50
Functions per module	5
Audit execution	Completed - AUDIT COMPLETE

Evidence was collected from local users and groups, Windows Security events, firewall profiles, BitLocker state, SMB configuration, services and Windows audit policy. Public repository screenshots have machine-specific identifiers removed where appropriate.

4. Key Findings

F-01 - Local administrative access | Medium

Observation: Two members were present in the local Administrators group. Review the membership against approved business need and least-privilege requirements. Recommendation: validate each member and remove unnecessary administrative access.

F-02 - Windows Firewall profiles disabled | High

Observation: Domain, Private and Public profiles were reported disabled with default policies not configured. Recommendation: verify whether another approved host firewall is protecting the endpoint; otherwise enable and baseline Windows Firewall.

F-03 - System volume not encrypted | High

Observation: The C: volume was reported FullyDecrypted with ProtectionStatus Off and 0% encryption. Recommendation: enable full-disk encryption where supported and establish recovery-key controls.

F-04 - SMB hardening gaps | High

Observation: SMBv1 was reported enabled, insecure guest logons were enabled and SMB signing was not required. Recommendation: disable SMBv1, remove insecure guest access and require signing where compatible with the environment.

F-05 - Windows audit-policy coverage requires review | Medium

Observation: multiple audit subcategories were reported as No Auditing. Recommendation: define and implement an approved Windows audit-policy baseline for required Success/Failure events.

F-06 - Endpoint protection collection limitation | Informational

Observation: Defender PowerShell/service interfaces were unavailable in the audit session. Recommendation: verify the installed endpoint security product and collect protection status through the supported management interface; do not treat the collection limitation as proof of a security failure.

Collection limitations recorded in the raw run included five Defender interface errors, two no-match event queries and one uptime conversion error. These should be handled as data-collection/implementation issues, not automatically promoted to security vulnerabilities.

5. Evidence and Reporting Outputs

Each execution creates a timestamped result set for review. The project separates source code, generated runtime output and recruiter-facing evidence so that machine-specific information is not unnecessarily published.

Artefact	Purpose
audit_report.csv	Machine-readable control results
audit_report.json	Structured result set for review or automation
summary.json	Assessment metadata and result counts
audit_report.html	Human-readable audit output
manifest.json	Collection/output manifest
audit-policy.txt	Windows audit-policy evidence
security-policy.inf	Exported local security-policy evidence

6. Reproducibility and Portfolio Presentation

A recruiter should be able to understand the project within seconds: **50 checks -> 10 modules -> evidence collection -> findings -> remediation**. The README therefore leads with the scope, technology and verified execution result, followed by a small evidence set and the complete report.

Public repository presentation should identify AI Shad Real Estate as a **fictional laboratory case study**. Do not describe it as client work. Do not claim ISO certification or organisational compliance based on this project.

Recommended recruiter sequence

Order	What the recruiter sees
1	README summary and project objective
2	10-module / 50-function validation screenshot
3	RBAC and Security Event Log evidence
4	Firewall / hardening evidence
5	Final findings and remediation
6	PDF report and source code

Portfolio claim: This project demonstrates practical PowerShell-based Windows security assessment and reporting in a controlled lab. It should be presented alongside, not instead of, commercial experience.